

Golf Launch Monitor Technology Review

**Measurement Principles, Patent Landscape, and Club/Ball
Parameter Estimation**

A technical foundation for the OpenFlight open-source launch monitor

OpenFlight Development

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Abstract

This review surveys the measurement technology behind commercial golf launch monitors — Doppler-radar systems (TrackMan, FlightScope, Garmin R10, Full Swing KIT), photometric camera systems (Foresight GCQuad/GC3, Uneekor, SkyTrak, ProTee VX, Garmin R50), and the hybrid radar+camera architectures that now dominate the high end — with emphasis on *how each system measures or infers club-delivery parameters* (club speed, club path, attack angle, face angle, dynamic loft, impact location) *and ball-launch parameters* (ball speed, launch angles, spin rate, spin axis). Primary sources include the governing patent families (TrackMan/Tuxen radar spin and spin-axis patents, the FlightScope/EDH dielectric-lens spin patent, the Wintriss/Foresight photometric patents, and the Acushnet stereo launch-monitor lineage), regulatory (FCC) filings, manufacturer technical documentation, and the peer-reviewed validation literature. The review develops the underlying physics — the D-plane impact model, oblique-impact spin generation, gear effect, and golf-ball aerodynamics — as the common mathematical core every launch monitor implements, and closes with a freedom-to-operate map of the patent landscape and concrete architectural recommendations for OpenFlight, an open-source Doppler-radar launch monitor built on the OmniPreSense OPS243-A and low-cost angle-radar modules. The central finding: every parameter a launch monitor reports sits on a *directness hierarchy* from measured to modeled, the industry has converged on optical augmentation of radar (and radar augmentation of cameras) precisely because neither modality can measure the full parameter set alone, and the recently expired Wintriss photometric patents (2025) open a practical low-cost optical path to true club-face and spin measurement that complements OpenFlight’s radar core.

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Chapter 1

Introduction

1.1 Purpose and scope

A golf launch monitor answers two questions for every shot: *what did the club do through impact* (delivery), and *what did the ball do after impact* (launch and flight). Commercial systems answer these questions with strikingly different sensor architectures — continuous-wave Doppler radar, multi-camera photogrammetry, and, increasingly, fusions of the two — yet they all implement the same underlying physics: an oblique-impact collision model connecting club delivery to ball launch, and an aerodynamic model connecting ball launch to flight.

This review is written to support the development of **OpenFlight**¹, an open-source launch monitor. Its goals are:

1. catalogue *how each major commercial system works*, at the level of radar physics, imaging geometry, and signal processing;
2. establish, parameter by parameter, *what is measured directly, what is inferred through a model, and what is essentially estimated* — especially for club-delivery data such as face angle and club path;
3. survey the governing *patent landscape*, since the strongest public documentation of these proprietary methods is found in the patents themselves, and since freedom-to-operate matters to an open-source project;
4. collect the *physics and algorithms* (D-plane, impact mechanics, gear effect, aerodynamic coefficients, spectral spin estimation, stereo triangulation, Kalman filtering) that any implementation needs; and
5. translate all of the above into *concrete design guidance* for OpenFlight’s radar-first architecture and its possible optical extensions.

1.2 The two sensing families

Doppler radar systems (TrackMan, FlightScope, Garmin Approach R10, Full Swing KIT) illuminate the hitting area and downrange volume with a microwave carrier and extract target radial velocity from the Doppler shift. Multiple receive antennas turn a velocity sensor into a 3D tracker via phase interferometry (Chapter 4). Radar excels outdoors: it tracks the entire flight, so carry and curvature are *observed*, not modeled. Its weaknesses are at the club face — radar cannot see face orientation directly — and indoors, where a screen truncates the observable flight.

Photometric (camera) systems (Foresight GC-series, Uneekor, SkyTrak, ProTee VX, Garmin R50) capture a burst of high-speed, IR-strobed stereo images over the first ~30 cm of ball flight

¹<https://github.com/jewbetcha/openflight> — AGPL-3.0; a DIY launch monitor built on the OmniPreSense OPS243-A 24 GHz Doppler radar, RFbeam K-LD7 angle radars, a sound-trigger board, and a Raspberry Pi 5.

and reconstruct position and orientation photogrammetrically (Chapter 5). They measure launch conditions — including 3D spin from dimple-pattern rotation — essentially perfectly for simulator purposes, then *model* the flight. Their weakness is the mirror image of radar’s: everything downrange of the capture volume is simulated, and club data requires either fiducial stickers on the face or an overhead viewing geometry.

The market’s convergent evolution is the single most instructive fact for a new design: *every* high-end vendor has concluded that neither modality suffices alone. TrackMan added cameras to its radar (OERT, Section 4.6); FlightScope added Fusion Tracking cameras; SkyTrak added radar to its camera; Rapsodo pairs radar with impact cameras; Full Swing feeds a camera into its radar ML pipeline. Chapter 6 tabulates the landscape.

1.3 Reading guide

Chapter 2 fixes terminology and coordinate conventions (TrackMan’s definitions, the de facto industry standard). Chapter 3 develops the impact physics connecting club delivery to ball launch — the D-plane model, spin generation, and gear effect — which every monitor uses either forward (simulation) or inverse (parameter estimation). Chapters 4 and 5 treat the two sensing families in depth. Chapter 6 surveys the commercial devices. Chapter 7 maps the patent landscape with expiry status. Chapter 8 covers ball-flight aerodynamics and trajectory estimation. Chapter 9 reviews the independent validation literature. Chapter 10 distills the implications for OpenFlight.

Key point

Throughout, we use a right-handed coordinate system for a right-handed golfer: x down the target line, y up, z to the golfer’s right. Angles are positive right/up. All club-delivery values are referenced to the moment of *maximum ball compression*; all ball-launch values to the moment of *separation from the face*.

Chapter 2

Parameter Definitions and Conventions

Launch monitors from different vendors disagree partly because they measure different things and partly because they *define* things differently. TrackMan’s definitions [31, 32] are the industry reference and are adopted here. The critical subtlety is *when* each quantity is defined: club-delivery parameters at the time of **maximum compression**, ball parameters **immediately after separation**.

2.1 Club-delivery parameters

Table 2.1: Club-delivery parameters (TrackMan conventions). “GC” = the geometric center of the club head.

Parameter	Definition	Notes
Club speed	Linear speed of the GC just prior to first contact	Not the impact-point speed; the toe moves up to ~ 7 mph faster than the heel
Attack angle	Vertical direction of GC motion at maximum compression	+ = hitting up. PGA Tour driver avg $\approx -1.3^\circ$; LPGA $\approx +3^\circ$
Club path	Horizontal direction of GC motion at maximum compression	+ = in-to-out (right of target for RH)
Face angle	Horizontal direction the face normal points, at the center-point of ball contact, at maximum compression	+ = open
Face to path	Face angle – club path	Sign controls curvature
Dynamic loft	Vertical angle of the face normal at the contact point at maximum compression	Differs from static loft via shaft lean/bend, face roll, impact height
Spin loft	3D angle between the club-motion direction (path, attack angle) and the face-normal direction (face angle, dynamic loft)	$\approx$ dynamic loft – attack angle only when face-to-path ≈ 0
Swing plane	Vertical angle of the plane traced by GC motion vs. the horizon	
Swing direction	Horizontal angle of that plane’s base vs. the target line	
Low point	Distance from GC at maximum compression to the lowest point of the swing arc	+ = low point ahead of the ball (ball-first contact)
Impact height / offset	Vertical / horizontal strike location relative to face center	Drives gear effect (Section 3.4)
Dynamic lie	Shaft angle vs. horizontal at impact	
Closure rate	Angular rate at which the face is closing in 3D ($^\circ/\text{s}$)	Foresight GCQuad-class only

Two definitional choices deserve emphasis because they are common sources of inter-device disagreement:

1. **Which point on the club is tracked.** TrackMan defines club speed at the *geometric center* of the head and reconstructs that point from the radar “3D silhouette” of the head [33]. A naive Doppler processor instead locks to the strongest or fastest return — often the toe of a driver — and reports a speed several mph high. This single choice explains much of the chronic club-speed disagreement between brands.
2. **Where on the face orientation is evaluated.** Face angle and dynamic loft are defined at the *contact point*, not the face center. On a curved driver face (bulge and roll, Section 3.4) the local normal at a toe strike differs from the center normal by several degrees, so systems that measure face pose but not impact location are systematically biased on off-center hits.

2.2 Ball-launch parameters

Table 2.2: Ball-launch parameters (defined immediately after separation).

Parameter	Definition
Ball speed	Speed of the ball’s center of gravity at separation
Smash factor	Ball speed $\div$ club speed
Launch angle	Vertical takeoff angle vs. the horizon
Launch direction	Horizontal takeoff angle vs. the target line
Spin rate	Rotation rate about the (single) spin axis, in rpm
Spin axis	Tilt of the rotation axis relative to the horizon; $-$ = tilted left $\Rightarrow$ draw for a right-hander
Carry / side / total	Trajectory descriptors; carry is measured to the point at launch elevation
Apex, landing angle	Peak height; descent angle at landing

A golf ball in flight has exactly one rotation vector $\boldsymbol{\omega}$. “Backspin” and “sidespin” are components of that vector, not separate spins. Systems that report sidespin (SkyTrak-style) and systems that report spin axis (TrackMan-style) are related by

$$S_{\text{side}} = S \sin \theta_{\text{axis}}, \quad S_{\text{back}} = S \cos \theta_{\text{axis}}, \quad \theta_{\text{axis}} = \text{atan2}(S_{\text{side}}, S_{\text{back}}). \quad (2.1)$$

A useful rule of thumb from TrackMan’s data: 1° of spin-axis tilt produces roughly 0.7% of carry as side-curve (about 0.7 yd per 100 yd) [61].

2.3 The directness hierarchy

For any launch monitor, each reported parameter falls somewhere on a hierarchy from directly measured to purely modeled. Anticipating the detailed treatment in Chapters 4 and 5, the hierarchy for the two sensing families is summarized in Table 2.3. This table is the skeleton of this whole review.

Table 2.3: Measured vs. derived, by architecture. “M” = measured directly, “D” = derived through a physical model, “E” = estimated/model-fit, “–” = not available.

Parameter	Radar (outdoor)	Radar (indoor)	Camera (photometric)	Hybrid
Ball speed	M	M	M	M
Launch angles	M	M	M	M
Spin rate	M ^a	M/E ^a	M ^b	M
Spin axis	D ^c	E	M ^b	M
Carry	M	E ^d	E ^d	E ^d
Club speed	M ^e	M ^e	M ^f	M
Club path / attack	M	M	M ^f	M
Face angle	D ^g	D ^g	M ^f	M
Dynamic loft	D ^g	D ^g	M ^f	M
Impact location	–	–	M ^f	M ^h

^a Doppler harmonic sidebands; requires surface asymmetry and sufficient flight (Section 4.4.2). ^b Dimple-pattern registration (Section 5.2.2). ^c Inverted from trajectory curvature via the Magnus constraint (Section 4.4.3). ^d Trajectory model integration from measured launch (Chapter 8). ^e Reference-point dependent; silhouette reconstruction on TrackMan. ^f Requires face fiducials (Foresight/Garmin R50) or overhead geometry (Uneekor/ProTee). ^g D-plane inversion from ball launch + path on radar-only units; optically assisted on OERT-class hardware (Section 4.5.2). ^h Markerless via camera on TrackMan 4/iO.

Chapter 3

Impact Physics: From Club Delivery to Ball Launch

Every launch monitor embeds a model of the club–ball collision. Camera systems use it *forward* (sanity-checking and filling gaps); radar systems use it *inverse* (recovering face angle and dynamic loft from measured ball launch). This chapter develops that model.

3.1 The D-plane

Jorgensen [1] observed that at impact two unit vectors determine the collision geometry for a center strike:

- $\hat{n}$, the **face normal**, built from face angle φ_f and dynamic loft λ_d ;
- $\hat{p}$, the **club-head velocity direction**, built from club path φ_p and attack angle α .

These two vectors span a wedge-shaped plane — the *D-plane* (“descriptive plane”). Its two governing consequences are:

1. **Launch direction lies in the D-plane**, between $\hat{n}$ and $\hat{p}$, much closer to the face normal.
2. **The spin axis is normal to the D-plane:**

$$\hat{\omega} \propto \hat{p} \times \hat{n}. \quad (3.1)$$

The ball therefore curves *away from the path, around the face* — the “new ball flight laws.”

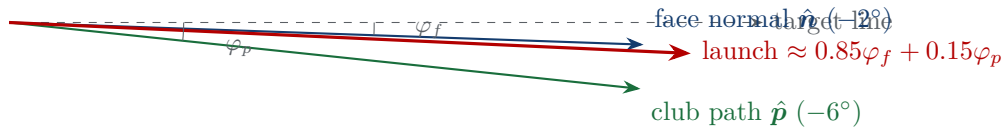


Figure 3.1: Horizontal D-plane geometry (top view, right-handed golfer, out-to-in “fade” delivery). The ball launches close to the face direction; the face-to-path difference tilts the spin axis and curves the ball away from the path.

3.1.1 Face/path weighting of launch direction

Robot and player data (TrackMan’s 2009 ball-flight-laws analysis [35]) give the canonical weighting of horizontal launch direction between face angle and club path:

$$\varphi_{\text{launch}} \approx w_f \varphi_f + (1 - w_f) \varphi_p, \quad w_f \approx \begin{cases} 0.85\text{--}0.87 & \text{driver} \\ 0.75 & \text{short irons / wedges.} \end{cases} \quad (3.2)$$

The weight is loft- and friction-dependent: more loft means a more oblique impact, hence a larger tangential (friction-driven) momentum component along the path direction. Peer-reviewed impact modeling with normal COR plus tangential compliance reproduces the 75–85% face dominance from first principles [10, 11].

Tutelman [58] gives a closed-form version calibrated to TrackMan data. With face-to-path angle A and dynamic loft L , define the total obliqueness Φ (which is precisely the spin loft):

$$\cos \Phi = \cos A \cos L, \quad f(\Phi) = 0.96 - 0.0071 \Phi \quad (\Phi \text{ in degrees}), \quad (3.3)$$

and the departure angles relative to the club path are

$$\text{DA}_{\text{vert}} = L f(\Phi), \quad \text{DA}_{\text{horiz}} = A f(\Phi), \quad (3.4)$$

with launch angles relative to the target obtained by adding attack angle and path respectively. Note $f(\Phi) \approx 0.85$ at $\Phi \approx 15^\circ$ (driver) and ≈ 0.75 at $\Phi \approx 30^\circ$ (short iron): one friction-calibrated function reproduces both rules of thumb.

3.1.2 Spin-axis tilt

For a center strike the spin-axis tilt follows from the D-plane geometry:

$$\tan \theta_{\text{axis}} \approx \frac{\tan(\text{face-to-path})}{\tan(\text{spin loft})}, \quad (3.5)$$

so 1° of face-to-path tilts the axis $\approx 4^\circ$ with a driver (spin loft $\sim 12\text{--}15^\circ$) but only $\approx 2^\circ$ with a mid-iron (spin loft $\sim 25\text{--}30^\circ$) [61]. Tuxen’s shot-shaping rule for a ball that curves back to the target line: $\theta_{\text{axis}} \approx -2.5 \times$ the horizontal launch angle.

Implication for OpenFlight

OpenFlight measures launch direction (K-LD7 horizontal) and club path. Equation (3.2) then yields face angle by inversion: $\varphi_f = (\varphi_{\text{launch}} - (1 - w_f)\varphi_p)/w_f$. Because $1/w_f \approx 1.15$, any systematic launch-direction bias is *amplified* $\sim 15\%$ in the reported face angle — alignment calibration of the horizontal angle radar is the single most important accuracy investment for club data.

3.2 Ball speed and smash factor

The normal-direction collision is well modeled as a two-body impact with coefficient of restitution e [2, 3]:

$$v_{\text{ball}} = v_{\text{club}} \frac{1 + e}{1 + m/M} \cos \Phi (1 - 0.14 x_{\text{miss}}), \quad (3.6)$$

with $e \approx 0.83$ at the USGA driver limit (falling with loft and impact speed), ball mass $m = 45.93$ g, head mass $M \approx 200$ g (driver), $\cos \Phi$ the obliqueness loss, and the last factor an empirical off-center loss with x_{miss} in inches [59]. The prefactor $(1 + e)/(1 + m/M) \approx 1.49$ sets the theoretical driver smash ceiling; smash factor falls with loft:

Effective loft	0°	10°	20°	30°
Max smash factor	1.488	1.465	1.398	1.288

Key point

This table is a built-in sanity check: a measured smash factor above the loft-appropriate ceiling indicates a measurement error — most commonly a radar unit reporting toe speed or shaft speed instead of geometric-center club speed. OpenFlight should flag physically impossible smash values instead of displaying them.

3.3 Spin generation

Friction during the oblique compression converts tangential surface speed $v_{\text{club}} \sin \Phi$ into rotation. Real impacts exhibit tangential compliance (the contact patch sticks and stretches like a shear spring) [10, 12], but a practical engineering fit calibrated to TrackMan data [58] is simply

$$S_{\text{total}} [\text{rpm}] \approx 160 v_{\text{club}} [\text{mph}] \sin \Phi, \quad (3.7)$$

with the back/side split following from the D-plane direction ratio $d = \tan A / \tan L$: $S_{\text{back}} = S / \sqrt{1 + d^2}$, $S_{\text{side}} = d S_{\text{back}}$. Cross-checks: $\sim 200\text{--}300$ rpm per degree of spin loft for a driver; the “iron loft $\times 200$ ” rule matches PGA Tour averages. Friction saturates near spin lofts of $45\text{--}50^\circ$ (the “spin-loft cliff”), and wet or grass-contaminated contact reduces spin at fixed spin loft [34].

3.4 Gear effect and impact location

An impact at horizontal offset x from the CG line torques the head about its CG; the recoiling face “gears” the ball the opposite way. The angular-impulse model [60] gives head rotation $\omega_{\text{head}} = x m v_{\text{ball}} / I_h$ and gear spin

$$s [\text{rpm}] = 58,830 \frac{v_{\text{ball}} [\text{mph}] C [\text{in}] x [\text{in}]}{I_h [\text{g cm}^2]} \approx 16.4 v_{\text{ball}} [\text{mph}] x [\text{in}] \times 100, \quad (3.8)$$

where C is CG depth behind the face (32–47 mm on measured OEM drivers) and $I_h = 4000\text{--}5800 \text{ g cm}^2$; the ratio I_h / C is nearly constant across modern drivers, which is why the simplified form works to $\pm 2.5\%$. A toe strike opens the head and imparts *hook* gear spin; a heel strike, slice spin. Vertical gear effect (high-face strikes reduce backspin, low-face strikes increase it) follows the same equation with the vertical CG offset and roll radius.

Driver faces are built curved to exploit this: **bulge** (horizontal radius, typically 12 in) starts a toe miss right so the gear-effect hook curves it back. Worked example [60]: a 1 in toe miss at 150 mph ball speed gets +1354 rpm of bulge-induced slice spin against -2192 rpm of gear hook spin — net 838 rpm hook and ~ 10 yd left, versus ~ 61 yd offline for a flat face. Sensitivity is extreme: a strike one dimple width (0.14 in) off-center tilts the spin axis $\sim 6^\circ$ on a driver ($\sim 2^\circ$ on a 6-iron) [61]. Irons have shallow CG depth, hence weak gear effect and flat faces.

Implication for OpenFlight

Gear effect is why *impact location* is the most valuable club parameter a launch monitor can add after path/face: without it, a pure D-plane inversion misattributes gear-effect spin-axis tilt to face-to-path. For a radar-only OpenFlight this is a fundamental, quantifiable error source on off-center driver strikes (up to several degrees of apparent face-to-path); Equation (3.8) bounds it, and an optical impact-location add-on (Chapter 10) removes it.

3.5 Vertical plane and the low-point geometry

The identical geometry applies vertically: launch angle sits 75–85% of the way from attack angle toward dynamic loft, and spin loft sets spin magnitude via Equation (3.7). One non-obvious coupling: for a descending strike on an inclined swing plane, the instantaneous path points in-to-out even when the swing direction is square — e.g. hitting 5° down on a 60° plane yields $\approx 2.9^\circ$ of in-to-out path. Radar systems that measure swing plane and attack angle use this relation ($\text{path} = f(\text{swing direction, swing plane, attack angle})$) as a consistency constraint among the measured club parameters.

Chapter 4

Doppler Radar Systems

4.1 Continuous-wave Doppler fundamentals

A CW radar transmits a carrier at frequency f_c (wavelength λ) and receives echoes shifted by the Doppler effect. For a target with radial velocity v_r ,

$$f_d = \frac{2v_r}{\lambda} = \frac{2f_c v_r}{c}. \quad (4.1)$$

At the 24.125 GHz K-band center frequency used by the OPS243-A (and the Garmin R10, Full Swing KIT, original Mevo), each 1 mph of radial speed produces ≈ 71.7 Hz of shift; at X-band (~ 10.5 GHz, TrackMan’s long-range subsystem and the Mevo+) the constant is ≈ 31.3 Hz/mph. A spinning, translating golf ball is not a point target: every surface patch has its own radial velocity, so the return is a velocity *spectrum* whose structure carries the spin information (Section 4.4.2).

Band choice is a range-vs-resolution trade. X-band supports long-range full-flight tracking (TrackMan tracks the entire ~ 6 s flight); 24 GHz gives finer velocity resolution per unit observation time and compact low-power hardware, but consumer 24 GHz units track only ~ 30 yd of flight [39, 40, 42].

4.2 From velocity sensor to 3D tracker: phase interferometry

A single-channel Doppler radar measures radial speed only. Every serious launch monitor adds *multiple receive antennas* and measures the *phase difference* of the return across receiver pairs (phase-comparison monopulse / interferometry). A wavefront arriving from direction $\hat{\mathbf{u}}$ reaches two antennas separated by baseline $\mathbf{d}$ with time delay $\tau = (\mathbf{d} \cdot \hat{\mathbf{u}})/c$, observed after mixing as a phase difference

$$\Delta\varphi = 2\pi f_c \tau \pmod{2\pi} \implies u = \frac{\lambda \Delta\varphi}{2\pi d}, \quad (4.2)$$

where u is the direction cosine along the baseline. Two orthogonal baselines give azimuth and elevation; the 2π ambiguities are resolved with more than three antennas at staggered spacings (TrackMan US 9,958,527 [15]). Combined with range (from multi-frequency CW phase differences or an FMCW chirp) and range rate (Doppler), the radar produces a full 3D track $(r, \text{az}, \text{el}, \dot{r})$ per epoch. Tuxen’s spin-axis patent describes exactly this “phase–phase monopulse comparison ... or interferometry” across three receivers [14]; the Garmin R10 uses a three-receiver 24 GHz array the same way [42]; FlightScope describes its aperture as a phased antenna array [41].

Implication for OpenFlight

OpenFlight’s architecture — OPS243-A for speed plus two K-LD7 modules for vertical and horizontal angles — is a physically separated version of what commercial units do on one board: the K-LD7’s RADC mode exposes raw ADC data from its two receive patches, and per-bin phase comparison Equation (4.2) yields the angle of each detection. The commercial

lesson is that angle accuracy is limited by baseline length, SNR, and mutual calibration; a rigid mechanical mount and a one-time angular calibration against surveyed targets matter more than any software refinement.

4.3 CW + FMCW hybrids

Full Swing’s patent application US 2020/0147470 [24] is the clearest public description of a hybrid launch-monitor radar: two ~ 24 GHz transmitters, time-division multiplexed, one CW and one linear-FMCW. The CW channel provides club speed, ball speed, spin, and launch-window angles from Doppler; the FMCW chirp provides direct range to 250 m+, carry, and side displacement from beat-frequency and phase measurements. The antenna layout is deliberately *non-uniform* — four receive antennas symmetric about the axis, two transmit antennas placed asymmetrically — to enrich the phase/frequency structure of the receive signal. FlightScope’s X3 similarly advertises “multi-frequency radar with direct distance measurement” [41].

4.4 Ball measurement

4.4.1 Speed and launch angles

Ball speed is read from the dominant Doppler line immediately after impact; launch angles come from the interferometric track over the first meters of flight. These are the most directly measured quantities in the whole parameter set and show the best inter-device agreement in every independent study (Chapter 9). Because the radar sits behind (or above) the ball, the measured radial speed underestimates the true speed by the cosine of the aspect angle; commercial units correct this using the tracked geometry, and the correction is largest for high launch angles and close radar placement.

4.4.2 Spin rate: the harmonic-sideband method

The foundational method is Tuxen’s patent US 8,845,442 [13] (EP 1 698 380). A golf ball’s dimples, paint, logo, seam, and core asymmetry make its radar cross-section vary periodically as it rotates. A surface feature at radius r contributes a Doppler term

$$f_{d,\text{feature}}(t) = \frac{2}{\lambda}(v_r - r\omega \sin \omega t), \quad (4.3)$$

i.e. frequency modulation of the return at the spin frequency $\omega/2\pi$ with deviation $(2/\lambda)r\omega$. The received spectrum therefore shows the main translational Doppler line flanked by **equally spaced harmonic sidebands** at

$$f = f_d \pm n f_{\text{spin}}, \quad n = 1, 2, 3, \dots \quad \implies \quad S [\text{rpm}] = 60 \Delta f_{\text{sideband}}. \quad (4.4)$$

The patented processing chain: (1) track the central velocity trace through the short-time Fourier transform; (2) detect symmetric sideband peaks; (3) track them over time as spectral traces; (4) *qualify* harmonics by verifying constant equal spacing across consecutive FFT frames; (5) solve the harmonic-number assignment; (6) divide trace offset by harmonic number [13, 36]. In practice a cepstrum or harmonic-product-spectrum estimator finds the comb spacing robustly.

Two practical corollaries: sideband SNR depends on ball-surface asymmetry (pristine two-piece range balls read weakly — the origin of “marked-ball” modes and Titleist RCT balls with embedded metal tags); and the comb yields spin *rate* only — the axis must come from elsewhere.

4.4.3 Spin axis: trajectory inversion

The same patent discloses the standard radar route to spin axis. From the differentiated 3D track, total acceleration $\mathbf{A}$ decomposes as $\mathbf{A} = \mathbf{g} + \mathbf{D} + \mathbf{L}$ with drag $\mathbf{D}$ antiparallel to the airspeed vector and Magnus lift $\mathbf{L}$ perpendicular to both the spin axis and the velocity. Projecting out gravity and drag,

$$\mathbf{D} = \left[\frac{(\mathbf{A} - \mathbf{g}) \cdot \mathbf{v}_a}{\|\mathbf{v}_a\|^2} \right] \mathbf{v}_a, \quad \mathbf{L} = \mathbf{A} - \mathbf{g} - \mathbf{D}, \quad (4.5)$$

and the orthogonality constraint $\mathbf{L} \cdot \hat{\boldsymbol{\omega}} = 0$, stacked over many trajectory points, gives an overdetermined linear system for the spin-axis direction $\hat{\boldsymbol{\omega}}$ [13]. This works well outdoors with seconds of observed flight; indoors, with 2–6 m of flight before the screen, the curvature signal is tiny and the axis becomes an estimate. FlightScope patented a direct alternative — time delays between vertically and horizontally separated receiver pairs yield the axis angle $\Phi = \arctan[S_H T_H / (S_V T_V)]$ [17] — precisely to recover the axis at short range.

4.4.4 Flight: tracked outdoors, modeled indoors

Outdoors, TrackMan-class units report *measured* carry, apex, and landing angle from the full track. Indoors every radar unit observes only the pre-screen segment (TrackMan 4 requires ≥ 4.7 m of throw; FlightScope claims 8 ft suffices with Fusion Tracking) and extrapolates with an aerodynamic model anchored to the vendor’s measured-flight database [38, 63]. The Garmin R10 illustrates the consumer floor: it requires ≥ 20 m of visible flight and ball speed above ~ 90 mph to measure spin at all; otherwise spin is estimated by a fitted model from the measured inputs [43].

4.5 Club measurement

4.5.1 What the radar tracks on the head

The clubhead is a large, complex reflector whose parts move at different speeds (toe faster than heel by up to ~ 7 mph on a driver). TrackMan defines club speed at the head’s *geometric center* and reconstructs that point by “directly measuring the 3D silhouette of the club head” [33]; naive processors lock onto the strongest or fastest return and systematically over-report. Attack angle and club path are the vertical and horizontal direction of the geometric center’s velocity at impact; swing plane, swing direction, and low point are properties of the center’s trajectory arc through the hitting zone.

4.5.2 Face angle and dynamic loft

Radar cannot see face orientation. On radar-only systems, face angle and dynamic loft are obtained by *inverting the D-plane collision model* (Section 3.1.1): given measured launch direction, club path, and the loft-dependent weighting w_f ,

$$\varphi_f = \frac{\varphi_{\text{launch}} - (1 - w_f) \varphi_p}{w_f}, \quad (4.6)$$

and analogously dynamic loft from launch angle and attack angle. TrackMan has confirmed these are “derived numbers from direct measurements and a collision model,” validated by robot testing [62]. On TrackMan 4/iO with OERT (Section 4.6), the company’s position is that club delivery including face angle at the contact point is measured via synchronized radar+camera silhouette tracking

rather than back-computed — though the exact split remains proprietary. Consumer units are explicit about the hierarchy: Garmin specifies launch direction at $\pm 1^\circ$ (measured) but face angle at $\pm 2^\circ$ and classifies it as calculated [43].

Key point

The reported “face angle” on a radar launch monitor is, to first order, a rescaled launch direction. It inherits launch-direction bias amplified by $1/w_f \approx 1.15$, plus a gear-effect error on off-center strikes that the model cannot see (Section 3.4). This is not a defect of any particular product but a structural property of the modality.

4.5.3 Known limitations

Independent benchmarking consistently finds club parameters far less accurate than ball parameters for radar (and for camera systems without face markers): the Leach et al. criterion study is the key reference [7]. Irons are harder than drivers (shaft and hosel returns, turf strike clutter, lower speeds); TrackMan’s advertised “>90% club-data pickup rate” with OERT is itself an acknowledgment that pickup is not universal [37]. Alignment error translates directly into correlated path/face bias on all units.

4.6 Optically enhanced radar: OERT and Fusion Tracking

TrackMan 4 fuses its dual radar (X-band long-range ball subsystem + 24 GHz high-resolution club/impact subsystem, receivers sampled at 40 ksps) with a built-in camera synchronized in time and space: the camera contributes exact pre-impact ball position, markerless *impact location* on the face, and “4D silhouette” clubhead tracking between radar epochs [37, 39]. The indoor-only TrackMan iO packs a 24 GHz radar with dual cameras — one at up to 4,600 fps for club and ball, one for alignment — under 810–850 nm IR illumination [38]. FlightScope’s patented equivalent is Fusion Tracking (US 10,338,209 [18]): the radar’s predicted track steers image search windows, the camera’s precise angular fixes correct the radar’s, and inter-sensor offsets are removed by error minimization. Rapsodo’s MLM2PRO pairs radar with 240 fps cameras that read the printed dot pattern on RPT/RCT balls for measured spin rate and axis [46]; Full Swing’s KIT feeds a 4K camera into an ML pipeline alongside its CW+FMCW radar [45].

Key point

Every major radar vendor has independently concluded that pure Doppler is insufficient at short range — for spin axis, impact location, and club-face data — and has added optical sensing or instrumented balls. This is the strongest architectural signal in the market for OpenFlight’s roadmap.

Chapter 5

Photometric (Camera) Systems

5.1 Imaging architectures

Photometric launch monitors capture a short burst of high-speed, IR-strobed stereo images over a small capture volume beside (or above) the tee — roughly the first 30 cm of ball flight — and reconstruct ball and club kinematics photogrammetrically. The commercial family tree:

- **Foresight Sports** defined the reference floor-unit design. The GC2 (2010) used two cameras (“stereoscopic”), with club data via the add-on HMT module (two more cameras). The GC3 / Bushnell Launch Pro (a Foresight-built rebadge) use three cameras; the GCQuad and QuadMAX use four cameras at the corners of the sensor window, capturing on the order of 200 images of ball and club per shot at burst rates quoted up to 10,000 fps; the ceiling-mounted GCHawk points the same quadrascopic package downward [47, 51]. Onboard IR object tracking detects the teed ball and the inbound clubhead and triggers the burst; intrinsics/extrinsics are factory-calibrated (“no calibration needed”) [47].
- **Uneekor** mounts overhead: EYE XO/XO2 look straight down from ~ 3 m with two (XO2: three) high-speed IR cameras at 3,000+ fps [52]. The overhead geometry sees the clubhead and face region directly, enabling sticker-free club data and impact-location video replay (Club Optix). The portable EYE MINI reverts to a side view and therefore requires club stickers.
- **ProTee VX**: overhead, two synchronized high-speed cameras with IR illumination and an ML shot-analysis pipeline; sticker-free club and ball data including vertical/horizontal impact point [56].
- **Garmin Approach R50**: three high-speed cameras arranged horizontally in a floor unit; purely optical ball and club measurement, with clubhead data requiring reflective fiducials [44].
- **SkyTrak / SkyTrak+**: the original SkyTrak was photometric and ball-only; the SkyTrak+ added a dual Doppler radar module specifically for club data while the cameras handle ball launch and spin [57].

5.2 Ball measurement

5.2.1 Photogrammetric position and velocity

With factory-known intrinsics K_i and extrinsics $[R_i|t_i]$, each synchronized image set yields a 3D ball-center fix by triangulation: back-project the matched detections as rays and solve the least-squares intersection (or the DLT system $\tilde{\mathbf{x}} \times (P\mathbf{X}) = 0$), refining by reprojection error. Ball speed is the displacement between timestamped fixes; launch angle and direction are the components of the initial velocity vector in the calibrated target frame. Two error sources dominate: depth error

grows quadratically with distance,

$$\sigma_Z = \frac{Z^2 \sigma_{px}}{f B}, \quad (5.1)$$

for baseline B and focal length f — which is why photometric units keep the capture volume small and also exploit the known ball diameter (21.34 mm radius) as an auxiliary range cue; and center-finding bias — fitting the projected *conic* outline rather than taking the blob centroid avoids a systematic offset of up to RZ/f pixels.

5.2.2 Spin: dimple-pattern registration

The flagship photometric capability is markerless 3D spin. Successive frames of the ball are registered on the sphere: the estimator searches the rotation $R \in SO(3)$ that best maps the back-projected surface texture (the dimple pattern, plus any logo or blemish) of frame k onto frame $k+1$ — Foresight brands this *spherical correlation*; Uneekor, *Dimple Optix* [50, 53]. The recovered R gives everything at once:

$$\hat{\omega} = \text{eig}_1(R), \quad \theta = \arccos\left(\frac{\text{tr } R - 1}{2}\right), \quad \omega = \theta/\Delta t, \quad (5.2)$$

i.e. the spin axis is the eigenvector of R with unit eigenvalue and the spin rate follows from the per-frame rotation angle. Reported accuracy is 1–3% of true spin, independent of ball speed. The known failure mode is rotational aliasing at high spin and low frame rate (the $n \cdot 2\pi$ ambiguity), resolved by multi-hypothesis tracking or a trajectory-consistency prior. Where resolution cannot support dimple registration, systems fall back to *marked balls*: Uneekor QED and Rapsodo (RPT balls) track printed high-contrast features, giving closed-form rotation from ≥ 3 correspondences via the Kabsch/Horn algorithm.

5.3 Club measurement

5.3.1 Fiducial markers: the Foresight approach

Foresight floor units require retroreflective **fiducial dots** on the clubface. The dot count maps directly to the data tier [48, 49]: one dot suffices for club speed, path, and attack angle (tracking a single 3D point through the capture volume); *four dots* — placed on the vertical centerline, equidistant from the horizontal centerline — define the face plane and enable face angle, dynamic loft, lie, closure rate ($^\circ/\text{s}$), and **impact location**. Mechanically: the IR strobe makes the dots bloom in every frame; triangulating the dot constellation across cameras and frames yields the 6-DOF pose of the face through impact. Club speed is the pose translation rate; path and attack angle are the velocity direction; face angle and dynamic loft are the face-plane orientation at contact; impact location is the measured ball position expressed in the face frame. The GC3/Launch Pro tier delivers speed, path, attack, face angle, and impact via the same stickers but omits dynamic loft, lie, and closure rate — these remain GCQuad/QuadMAX exclusives [51].

5.3.2 Sticker-free overhead tracking

Uneekor’s overhead cameras segment the clubhead silhouette directly — no markers — because the top-down view presents the head against the mat with controlled IR illumination; the systems also record actual high-speed impact video (face-on via Club Optix), from which impact location is read [52]. ProTee VX claims the same sticker-free club set (speed, path, face, attack, dynamic loft, lie, vertical and horizontal impact point) from its overhead ML pipeline [56]. The trade-off is installation: overhead units need $\sim 3\text{ m}$ ceilings, rigid mounting, and a floor-chart calibration.

5.3.3 Calibration

Factory-rigid multi-camera rigs ship calibrated; the user aligns only the target line. Overhead systems require field calibration: Uneekor’s procedure places a printed calibration chart on the mat, levels it, squares it to the screen, and aligns its crosshair to overlays on the live camera feeds — establishing the ground plane, hitting zone, and target-line extrinsics [55]. This is the pattern a DIY overhead build should copy.

5.4 The indoor argument, and what cameras cannot do

Photometric systems measure actual launch conditions — including 3D spin — in the first foot of flight; nothing is lost when the ball hits a screen 2.5 m away. Radar indoors must infer from a truncated flight. This is the structural reason camera units dominate indoor simulation while radar dominates outdoor practice: each modality measures what the other models [54]. The camera’s blind spots are downrange (all flight is simulated from launch) and, for floor units, the club face itself without stickers — the same face-visibility problem radar has, solved with markers instead of models.

5.5 DIY and open-source photometric systems

PiTrac [66] is the flagship open-source photometric build and the most relevant external project to OpenFlight. Instead of multi-kfps cameras it uses *strobed multi-exposure capture*: an IR LED array pulses several times within one long exposure of a ~\$50 Raspberry Pi Global Shutter camera, freezing multiple ball images in a single frame. One camera watches the teed ball and detects launch; the second captures the strobed flight images; a custom PCB sequences trigger and strobe. Processing (C++/OpenCV): Hough-circle detection locates ball images; strobe-timestamped positions give speed and launch angles; 3D spin on all three axes is solved by searching candidate rotations that best register the dimple imagery between exposures — Gabor-filter dimple enhancement is documented in the adjacent patent literature [26]. Total BOM ~\$250–300. The architecture is a direct descendant of the now-expired Wintriss patents (Section 7.2) and of the GC2’s flash-multi-exposure technique.

Implication for OpenFlight

PiTrac proves that commodity global-shutter sensors + IR strobing deliver photometric-grade launch and spin measurement at OpenFlight’s price point. For OpenFlight the natural division of labor is: keep the radar core for speed/trigger robustness and outdoor use, and adopt a PiTrac-style strobed camera as the spin/impact-location module indoors — the same fusion direction every commercial vendor has taken, from the opposite starting modality.

Chapter 6

Commercial System Survey

Table 6.1 summarizes the sensing architecture of the major systems; the notes that follow give the technically salient details per device, with sources in the bibliography.

Table 6.1: Launch monitor architecture survey (2026). “Club source” distinguishes directly measured face data (optical) from D-plane-inverted or model-estimated face data.

System	Primary sensing	Spin method	Club data source	Impact location
TrackMan 4	Dual radar (X-band + 24 GHz) + camera (OERT)	Doppler sidebands	Radar silhouette + camera	Yes (markerless, OERT)
TrackMan iO	24 GHz radar + 4,600 fps camera	Camera + radar	Radar + camera	Yes (markerless)
FlightScope X3	Phased-array multi-freq. radar + Fusion cameras	Doppler (dielectric-lens)	Radar; face derived	No
FlightScope Mevo+	X-band 10.5 GHz phased array (+ Pro cameras)	Doppler; stickers indoors	Radar; face derived	No
Garmin R10	24 GHz, 3-receiver CW	Doppler (≥ 20 m flight) or model	Radar; face calculated	No
Garmin R50	3 high-speed cameras	Dimple imaging	Optical (fiducials)	Yes
Full Swing KIT	24 GHz CW+FMCW + 4K camera ML	Doppler + ML	Radar + ML	No
Rapsodo MLM2PRO	Radar + 2×240 fps cameras	Marked ball (RPT/RCT)	Radar; face derived	Video only
Foresight GCQuad / QuadMAX	4 cameras, IR strobe	Spherical correlation	Optical (4 fiducials)	Yes (measured)
GC3 / Launch Pro	3 cameras, IR strobe	Spherical correlation	Optical (fiducials; add-on)	Yes
GCHawk	4 cameras, ceiling	Spherical correlation	Optical	Yes
Uneekor EYE XO2	3 overhead IR cameras, 3,000+ fps	Dimple Optix	Optical, sticker-free	Yes + video
Uneekor EYE MINI	2 cameras, portable	Dimple Optix	Optical (stickers)	Yes
ProTee VX	2 overhead cameras + ML	Dimple imaging	Optical, sticker-free	Yes
SkyTrak+ / ST MAX	Cameras (ball) + dual radar (club)	Photometric	Radar; face derived	No
OpenFlight (current)	OPS243-A 24 GHz + 2×K-LD7 + sound trigger	Doppler I/Q buffer (~ 50 – 60% detect)	Radar; no face data yet	No
PiTrac (DIY)	2 Pi GS cameras + IR strobe	Dimple registration	None yet	No

6.1 Radar-first systems

TrackMan 4 / iO. The reference radar architecture: two synchronized radar subsystems (long-range X-band at the corners for full ball flight; higher-frequency 24 GHz at the center for club and impact), receivers sampled at 40 ksps to pin the impact instant, fused with an OERT camera [39, 37, 38]. Reports 27+ parameters including swing plane/direction, low point, dynamic lie, and D-plane-derived face data; requires ≥ 4.7 m of indoor throw. The iO is the indoor-optimized repackaging: 24 GHz radar, 4,600 fps club/ball camera, IR illumination, no minimum-space requirement.

FlightScope X3 / Mevo+. Phased-array Doppler (X3: multi-frequency with direct ranging; Mevo+: X-band 10.5–10.55 GHz per FCC filings [40]) with Fusion Tracking cameras on the X3 and Mevo+ Pro. Spin measured via the patented dielectric-lens phase-demodulation route (Section 7.3); metallic stickers recommended for short indoor flights. FlightScope’s 2022 win against TrackMan at the German Federal Court of Justice, after losing the 2013 EP 1 698 380 case, bookends two decades of radar-spin litigation between the two [30].

Garmin Approach R10. The instructive budget case: a 24 GHz three-receiver CW radar that directly measures only the safe primitives — ball speed, launch angles, club speed, club path — and models the rest. Spin requires ≥ 20 m of flight and >90 mph ball speed (or an RCT marked ball indoors); face angle is explicitly a calculated value at $\pm 2^\circ$ [43, 42]. Garmin’s step-up R50 abandons radar entirely for three cameras — a telling modality switch for indoor accuracy.

Full Swing KIT. The patented CW+FMCW dual-mode 24 GHz radar (Section 4.3) with an ML vision assist from its 4K camera; 16 parameters; third-party testing places it within 1–2% of TrackMan/GCQuad in most conditions [45].

Rapsodo MLM2PRO. Radar for speed/launch plus two 240 fps cameras that read printed RPT/RCT ball markers for measured spin rate and axis (claimed within 1% of reference units) — the cheapest route to *measured* spin, at the cost of proprietary balls [46].

6.2 Camera-first systems

Foresight GCQuad / QuadMAX / GCHawk. Quadrascopic IR-strobed capture (~ 200 images/shot); spherical-correlation spin; four-dot fiducial club measurement with closure rate and measured impact location (Section 5.3) [47, 48]. Robot testing shows the class-leading spin repeatability (Chapter 9).

GC3 / Bushnell Launch Pro. Identical triscopic hardware in two brands; club data is a paid add-on tier using the same stickers [51].

Uneekor EYE XO2 / EYE MINI / ProTee VX. Overhead sticker-free club measurement (XO2, VX) versus portable sticker-based (EYE MINI); Dimple Optix markerless spin; impact video replay [52, 56].

SkyTrak+ / ST MAX. The camera-to-radar convergence case: the original photometric ball-only SkyTrak gained a dual-radar club module and ML fusion in the SkyTrak+ [57].

6.3 Open-source systems

OpenFlight (radar-first: OPS243-A I/Q rolling buffer + sound trigger + K-LD7 interferometric angle radars) and **PiTrac** (camera-first: strobed multi-exposure global-shutter imaging) occupy the two ends of the same spectrum the commercial market spans, at roughly 1/10th the hardware cost. Neither yet measures club face data; both have clear, patent-informed paths to it (Chapter 10).

Chapter 7

Patent Landscape

The strongest public documentation of proprietary launch-monitor methods is the patents themselves; they are cited throughout this review as technical sources. This chapter organizes them by assignee and closes with a freedom-to-operate (FTO) map. Expiry estimates follow the 20-years-from-priority rule plus patent-term adjustment (PTA), using Google Patents’ anticipated-expiration data where shown; *none of this is legal advice, and claim-by-claim analysis by counsel is required before any commercial decision.*

7.1 TrackMan A/S (Interactive Sports Games A/S) — Fredrik Tuxen

TrackMan’s public patent list enumerates 45 US patents covering full-trajectory radar tracking, club delivery data, radar spin rate and spin axis, markerless impact location, and OERT fusion [27].

US8845442B2 — “Determination of spin parameters of a sports ball.” Priority March 3, 2005; PTA-extended expiry ~May 2029 [13]. *The* radar-spin patent: harmonic sidebands equally spaced at the spin frequency around the central Doppler line, with the trace-tracking/qualification chain of Section 4.4.2; spin axis from the Magnus-lift orthogonality constraint (Equation (4.5)). The European sibling EP 1 698 380 was upheld by Germany’s Federal Court of Justice and grounded TrackMan’s 2013 Düsseldorf win against FlightScope’s distributor. Continuation US10393870B2 (same disclosure) expires December 2026.

US8085188B2 / US9857459B2 / US10473778B2 — target-deviation family. Priority July 2004. Camera rigidly mounted to the radar; user taps a target in the image; system reports launch-to-target deviation with automatic coordinate transforms. US9857459 lapsed 2022; the siblings expire 2026–27 — a safe UX pattern to adopt shortly.

US10850179B2 (+ US11446546, US11938375) — spin axis. Direct spectral/interferometric spin-axis determination from per-receiver-pair phase differences; also documents the three-receiver monopulse architecture [14].

US10596416 / US11697046 / US12128275 — radar+image 3D tracking. The OERT family; active into the late 2030s.

US10315093B2 — trajectory illustration. The broadcast “tracer” overlay (radar track rendered into calibrated video); to 2030.

US11086005B2 — multi-bay tracking. Toptracer-style back-extrapolation of tracks to assign shots to bays; to ~2036.

7.2 Foresight Sports (WAWGD) and the Wintriss lineage

Foresight Sports is WAWGD, Inc.; the foundational photometric patents it holds (via Wawgd Newco LLC) were invented at Wintriss Engineering by Christopher Kiraly (a Foresight co-founder) — and these same numbers appear on Uneekor’s license list following the September 2024 Foresight–Uneekor license agreement [29].

US7292711B2 — “Flight parameter measurement system.” Priority June 6, 2002; **expired April 2025** [19]. The blueprint single-camera photometric monitor: factory per-pixel 3D calibration; accelerometer leveling; microphone + small radar horn joint trigger; strobe-lit sequential images; ball center/diameter with known ball size giving 3D position per frame; and *markerless spin* by iterative rotation/correlation of natural surface features (dimples, blemishes) after glint removal and lighting normalization.

US7324663B2 — sibling “smart camera” patent. Same spec, self-triggering from in-FOV motion; **expired August 2025.**

US7497780B2 / US7641565B2 — integrated monitor and ball-placement detection. Priority 2006; expire ~2027. The GC2-style UX: optical ball-find, LED placement guidance, frame-differencing launch detection, mixed-mode capture, on-device display. Foresight enforced this portable-monitor family as recently as 2024 (Uneekor settlement).

Later WAWGD filings. Applications on measuring club path and face orientation before/at/after impact — the four-dot GCQuad fiducial system — remain active; treat quad-camera + reflective-dot face measurement as protected.

7.3 FlightScope / EDH — Henri Johnson

US9868044B2 — “Ball spin rate measurement.” Priority January 2013; expires ~2034 [16]. The engineered alternative to TrackMan’s sidebands after the 2013 loss: the ball’s cover acts as a *dielectric lens* ($n \approx 1.8$) magnifying far-side surface features in the microwave return; phase demodulation (PLL) yields repeating bipolar pulses as features sweep the magnification zone; an FFT extracts the periodicity (seam symmetry doubles the modulation rate, corrected in software).

US10775492B2 — “Golf ball spin axis measurement.” Priority December 2013; expires ~2035 [17]. Direct axis measurement from time delays between perpendicular receiver pairs: $\Phi = \arctan[S_H T_H / (S_V T_V)]$ — works at short indoor flights where trajectory inversion fails.

US10338209B2 — Fusion Tracking. Priority 2015; expires ~2037 [18]. Radar+camera fusion with checkerboard camera calibration, Doppler-simulator radar alignment, radar-steered image search windows, offset removal by error minimization.

7.4 Acushnet (Titleist) — the stereo foundation

The deepest prior art for camera-based monitors; the early family is entirely expired [20, 21, 22].

US5501463A (1992, expired). Two shuttered cameras at $\sim 22^\circ$, double-strobed $\sim 800 \mu\text{s}$ apart; three retroreflective dots on the clubhead, six on the ball; triangulation yields 3D clubhead velocity, attack angle, path, face orientation, and *contact location on the face*.

US6500073B1 (1992 priority, expired). Stereo pair + sound trigger + six ball dots; position and orientation at two instants give velocity and angular velocity; numerical flight integration (drag/Magnus/gravity) gives carry and roll — the classic tour package.

US6758759B2 (2001, expired 2022). Dual two-camera monitors (club pre-impact, ball post-impact); magnetic fixture calibrates the head’s geometric center; complete recipe for measured face angle and impact location with marker stickers.

US7143639B2 family (2004 priority; parent expired 2024). Portable four-camera unit with speed-adaptive strobe timing via FPGA lookup, dichroic marker discrimination, optical club/ball fingerprinting; continuations US8500568B2/US8556267B2 (hardware integration claims) run to ~2030–31.

US10668350B2 (2017; to ~2038). “True 3D” stereo/light-field capture at 1,000–10,000+ fps with sub-10 μ s exposures and per-frame xyz measurement.

US6186002B1. USGA-adjacent method for extracting C_D/C_L from measured trajectories — a template for calibrating an open ball-flight model [23].

7.5 Korean ecosystem: Creatz/Uneekor and Golfzon

Uneekor’s patent list mixes owned Creatz patents (US10247553, US9752875, US9605960, US9448067, US10587797, US10776929, US11191998, US12008770 — including the Dimple Optix markerless-spin engine) with the licensed Foresight/Wintriss numbers [28]. US10247553B2 (Creatz; to ~2032) claims a sectioned start-sensor detecting the actual ball starting position for simulation. Golfzon’s US9242158B2 (to ~2032) claims the latency-hiding two-stage pipeline — start the simulated trajectory from fast “first ball information” (~100 ms), refine in flight when the slower spin estimate arrives (~200 ms) [25].

7.6 Freedom-to-operate map

Key point

Three strategic conclusions. (1) A *camera-first* open design has a wide-open, recently expired foundation (Wintriss + Acushnet) covering mono/stereo photometric launch measurement, markerless spin, and marker-based measured club-face data. (2) *Radar spin* is the most encumbered corner: the TrackMan sideband family runs to ~2029 and FlightScope’s alternatives to ~2034–35; pure speed/launch-angle radar (the Garmin R10 recipe) is safe. (3) *Radar+camera fusion* is the most actively patented current frontier — the two giants have litigated each other in both directions — and warrants the most care for any hybrid OpenFlight roadmap. Note also that a non-commercial AGPL project is not immune: US patent infringement does not require sale, and downstream commercial users inherit the exposure.

Table 7.1: FTO summary for an open-source launch monitor (US perspective, mid-2026). Verify claim-by-claim with counsel before commercial use.

Technique	Controlling patents	Status
CW Doppler speed measurement	—	Ancient art; free
Mono-camera photometric launch + markerless dimple-correlation spin	US7292711 / US7324663	Expired 2025; free
Stereo + retroreflective dots: club face, path, impact location	US5501463 / US6500073 / US6758759 / US7143639	Expired; free
Target-tap deviation UX	US8085188 family	Expires 2026–27
Radar spin via harmonic sidebands	US8845442 (US10393870)	To ~2029 (Dec 2026)
Radar spin via dielectric-lens phase demodulation	US9868044	To ~2034
Direct radar spin axis (perpendicular receiver pairs)	US10775492	To ~2035
Radar+camera fusion tracking	US10338209; Track-Man OERT family	To ~2037+
Integrated ball-find/placement/trigger workflow	US7497780 / US7641565	To ~2027
Video tracer overlay from sensor track	US10315093	To ~2030
Two-stage sim latency hiding	US9242158	To ~2032
Start-position sensing	US10247553	To ~2032
Light-field / true-3D capture claims	US10668350	To ~2038
Multi-bay back-extrapolation attribution	US11086005	To ~2036

Chapter 8

Ball Flight Models and Trajectory Estimation

Both sensing families depend on an aerodynamic model: cameras integrate it forward from measured launch to get carry; radars fit it to partial trajectories to extract spin axis and to extrapolate indoor flights.

8.1 Equations of motion

With air density ρ , ball radius $R = 21.34\text{ mm}$, mass $m = 45.93\text{ g}$, cross-section $A = \pi R^2$, and airspeed $\mathbf{v}_a = \mathbf{v} - \mathbf{v}_{\text{wind}}$:

$$m \frac{d\mathbf{v}}{dt} = -\frac{1}{2}\rho AC_D \|\mathbf{v}_a\| \mathbf{v}_a + \frac{1}{2}\rho AC_L \|\mathbf{v}_a\|^2 (\hat{\boldsymbol{\omega}} \times \hat{\mathbf{v}}_a) + m\mathbf{g}, \quad (8.1)$$

with the drag and lift coefficients functions of Reynolds number $\text{Re} = 2R\|\mathbf{v}_a\|/\nu$ and spin ratio $S = R\omega/\|\mathbf{v}_a\|$. Fourth-order Runge–Kutta integration is ample; carry is evaluated where the trajectory returns to launch elevation. Tilting the spin axis by θ rotates the Magnus force off vertical, giving lateral acceleration $(F_M/m)\sin\theta$ — the mechanism behind the 0.7%-per-degree side-curve rule and Equation (2.1).

8.2 Aerodynamic coefficient data

Bearman & Harvey (1976) [4]. The canonical wind-tunnel dataset on spinning golf-ball models across the flight envelope. Dimples drop the critical Reynolds number to $\sim 5 \times 10^4$; post-critical $C_D \approx 0.25$ is nearly Re-independent; C_L rises with spin ratio from ≈ 0.08 to 0.25 over $S \approx 0.02$ –0.3; hexagonal dimples outperform round ones.

Smits & Smith (1994) [5]. The parameterized model most widely used in simulators:

$$C_D = C_{D1} + C_{D2} S + C_{D3} \sin(\pi \frac{\text{Re} - A_1}{A_2}), \quad C_L = C_{L1} S^{0.4}, \quad (8.2)$$

with $C_{D1} = 0.24$, $C_{D2} = 0.18$, $C_{D3} = 0.06$, $A_1 = 9 \times 10^4$, $A_2 = 2 \times 10^5$, $C_{L1} \approx 0.54$, plus a spin-decay law with time constant $\approx 24\text{ s}$ at 100 mph (roughly 4%/s early in flight).

Quintavalla / USGA Indoor Test Range (2002) [6]. Six-term polynomial C_D/C_L models in Re and S fitted from trajectory photography on the USGA ITR; captures the low-speed end-of-flight regime better than wind tunnels; the basis of USGA Overall Distance Standard conformance testing. The companion method patent US6186002 [23] — determining coefficients from measured trajectories — is the template for calibrating an open model against real flights.

Implication for OpenFlight

OpenFlight should ship Smits–Smith (Equation (8.2)) with spin decay as the default flight model, structured so a Quintavalla-style six-term fit can drop in, and calibrate against measured outdoor trajectories (or MLM2PRO reference data) using the US6186002 trajectory-fitting approach. Carry disagreements between simulators are dominated by model choice, not launch measurement — version the model and record it with every shot log.

8.3 Trajectory estimation and filtering

The estimation core of a radar launch monitor is a nonlinear filtering problem. State $\mathbf{x} = (\mathbf{p}, \mathbf{v}, S, \theta_{\text{axis}})$; process model = Equation (8.1); measurement models: radar $h(\mathbf{x}) = (r, \text{az}, \text{el}, \dot{r})$ with $\dot{r} = \mathbf{v} \cdot \hat{\mathbf{p}}$, camera $h(\mathbf{x}) = \text{pixel projections}$. An extended (or unscented) Kalman filter runs the standard recursion; launch parameters are then obtained by *smoothing and back-extrapolation* — fit the entire observed arc and evaluate the state at the moment of face separation, which is far more robust than differencing the first noisy fixes. Including θ_{axis} in the state makes spin-axis-from-curvature (Section 4.4.3) fall out of the filter naturally, and per-measurement confidence weighting is the clean way to fuse heterogeneous sensors (OPS243-A speed, K-LD7 angles, future camera fixes).

Implication for OpenFlight

OpenFlight currently correlates K-LD7 angle bursts with the OPS243-A impact timestamp and reads angles from single detections. Migrating to a short EKF smoother over the full K-LD7 ring-buffer burst — even 10–20 detections over 50 ms — would use all available data, reject outlier bins, and yield launch angles with quantified covariance. The same filter skeleton later absorbs camera measurements unchanged.

8.4 Indoor extrapolation and its errors

Indoors, every system integrates Equation (8.1) from launch conditions. Error propagation is dominated by spin uncertainty: at driver speeds, ± 300 rpm maps to roughly ± 4 – 6 yd of carry and ± 1 yd of side; a $\pm 2^\circ$ spin-axis error at 2,500 rpm maps to ± 3 – 4 yd of side at 250 yd carry. This is why the accuracy literature (Chapter 9) finds spin to be the fragile channel indoors for radar, and why camera systems — which measure spin directly — win the indoor comparison structurally.

Chapter 9

Accuracy: The Independent Evidence

9.1 Peer-reviewed validation

Leach, Forrester, Mears & Roberts (2017) [7] remains the key criterion study: 240 shots (driver, 7-iron, wedge) measured simultaneously by a TrackMan Pro IIIe, a Foresight GC2+HMT, and a four-camera 5,400 fps optical reference. Findings: ball parameters agreed well on both devices (TrackMan clubhead-speed median difference -0.4 mph; ball speed $+0.2$ mph; launch angle 0.0°), but **club parameters were materially worse for both** — the authors endorse ball data for research use and advise caution on club data. This is the empirical face of the directness hierarchy of Section 2.3: what is measured agrees; what is derived diverges.

TrackMan 4 indoor reliability (J. Sports Sciences, 2024) [8]: within- and between-session reliability in high-level golfers indoors — ICC 0.99 for club speed, 0.97–0.99 for ball speed, but **spin-rate ICC as low as 0.02–0.60**: even the reference radar’s indoor spin channel is fragile when flight is truncated.

A 2025 Mevo+ vs. TrackMan 4 indoor agreement study exists [9] (abstract-level access only at review time), extending the same pattern down-market.

9.2 Robot and industry testing

Golf Laboratories robot testing (reported via Foresight and third parties) puts GCQuad center-strike spin standard deviation at ≈ 82 rpm versus ≈ 175 rpm for TrackMan 4 — the photometric spin advantage in its clearest form [64]. MyGolfSpy’s indoor comparison against a GCQuad reference found the camera-assisted MLM2PRO among the tightest budget units for launch, spin, and carry, with radar-only units (R10, original Mevo) trailing indoors [65]. Manufacturer-quoted specs bracket the market: ± 1 mph ball speed and $\pm 1^\circ$ launch angles are common claims; club-face claims ($\pm 2^\circ$, “calculated”) are visibly weaker.

9.3 What drives inter-device disagreement

Synthesizing the validation literature and the architecture analysis:

1. **Reference-point differences** in club speed (geometric center vs. fastest return) — several mph of systematic spread.
2. **Derived face data**: D-plane inversion amplifies launch-direction bias by $\sim 1.15\times$ and cannot see gear effect, so radar face angle diverges from optical face angle most on off-center strikes.
3. **Indoor spin**: sideband SNR collapses on short flights and clean balls; systems silently switch to estimation, and estimated spin feeds the carry model.

4. **Flight-model differences:** identical launch conditions produce different simulated carries across vendors; this is a modeling disagreement, not a measurement one.
5. **Alignment:** a unit misaligned to the target line biases path, face, and launch direction coherently — the cheapest error to fix and the most common in practice.

Implication for OpenFlight

For OpenFlight validation against the MLM2PRO (the project’s reference instrument): compare ball speed and launch angles directly; compare spin only on RPT/RCT marked balls where the MLM2PRO’s measurement is camera-based; expect club-speed offsets from reference-point differences and calibrate a per-club correction rather than chasing agreement; and log raw I/Q captures so spin-detection improvements can be replayed against historical shots.

Chapter 10

Implications for OpenFlight

OpenFlight’s current architecture — OPS243-A 24 GHz CW Doppler with rolling-buffer I/Q capture, a hardware sound trigger, and two K-LD7 interferometric angle radars, on a Raspberry Pi 5 — is a minimal but honest instance of the commercial radar architecture. This chapter distills the review into design guidance, ordered by leverage.

10.1 Ball data

1. **Ball speed** is already the strongest channel: the Doppler line is unambiguous, and the OPS243-A’s $\pm 0.5\%$ class accuracy matches commercial practice. Add the aspect-angle (cosine) correction using the measured launch angles — at 3–5 ft behind the tee with a driver launch of 12° the radial underestimate is small but systematic.
2. **Launch angles**: treat the K-LD7 bursts as a short trajectory, not a single detection — an EKF smoother over the ring buffer (Section 8.3) with back-extrapolation to the impact timestamp is the single highest-leverage software upgrade for angle quality.
3. **Spin rate**: the current $\sim 50\text{--}60\%$ detection rate from I/Q sideband analysis is consistent with the physics: sideband SNR depends on ball-surface asymmetry and observation time (Section 4.4.2). Improvements in order of cost: longer observation windows (later trigger cutoff); cepstrum/harmonic-product comb estimation with the patent-style equal-spacing qualification; logo-ball or marked-ball guidance for indoor use; and honest fallback — report estimated spin (from club type + speed + loft priors, clearly flagged) when no comb is found, as Garmin does. *Patent caution*: the harmonic-sideband method is claimed by US8845442 until ~ 2029 (US member); for a hobby project this is a risk-management judgment, but the roadmap should note the family’s expiry dates (Section 7.6).
4. **Spin axis** is currently out of reach of the radar: indoor flights are too short for trajectory inversion, and the direct receiver-pair method is FlightScope-patented to ~ 2035 . The practical route is optical (below) or a flagged model estimate from face-to-path.

10.2 Club data

1. **Club speed**: define and document the reference point. The OPS243-A sees a smear of head/shaft returns; gating the pre-impact spectrum and taking a fixed percentile of the velocity distribution (rather than the maximum) approximates a stable reference and avoids the toe-speed inflation that plagues naive processors (Section 4.5). Validate the smash factor against the loft-appropriate ceiling (Section 3.2) and flag violations instead of displaying them.
2. **Club path** (K-LD7 horizontal) and **attack angle** (vertical, if geometry allows) are legitimate direct measurements — the same primitives the Garmin R10 measures. Alignment calibration dominates their accuracy.

3. **Face angle:** implement the D-plane inversion (Equation (4.6)) with the loft-dependent weight (Equation (3.3)) — this is exactly what radar-only commercial units report — but label it *derived*, propagate its error ($1.15\times$ launch-direction bias), and suppress it on detected off-center strikes once impact location is available. The D-plane forward model doubles as the simulator’s launch generator, so one well-tested module serves both directions.
4. **Impact location** requires optics. The expired Acushnet stereo art (Section 7.4) and Wintriss mono-camera art (Section 7.2) provide complete public-domain recipes; a PiTrac-style Pi Global Shutter camera with IR strobing is the natural hardware. Even a single overhead camera reading face stickers (Acushnet US6758759 recipe, expired) would convert OpenFlight’s face angle from derived to measured.

10.3 Architecture roadmap

1. **Phase 1 (radar-only, current):** EKF smoothing, comb-based spin with honest fallback, D-plane-derived face data with flags, smash-factor sanity gates, alignment-calibration procedure. Ships the Garmin-R10 feature set with open data.
2. **Phase 2 (optical spin/impact module):** one or two Pi Global Shutter cameras + IR strobe PCB (PiTrac-proven, $\sim \$100$ incremental). Camera measures launch angles, 3D spin (dimple registration, public-domain per expired US7292711), and impact location; radar keeps trigger, speed, and outdoor robustness. This is the same fusion every commercial vendor converged on — but built from the expired-art side of the patent map. *Care:* avoid the still-active claims around integrated ball-find/LED-guidance UX (to ~ 2027) and radar-steered image search windows (FlightScope US10338209).
3. **Phase 3 (fusion):** a single EKF consuming OPS243-A speed, K-LD7 angles, and camera fixes, with per-sensor covariances; flight model per Chapter 8 with versioned coefficients; validation protocol against the MLM2PRO per Chapter 9.

10.4 Reporting honesty as a feature

The single clearest lesson of the review: commercial marketing blurs the measured/derived boundary, and the validation literature repeatedly punishes the derived quantities. An open-source instrument can win trust by doing the opposite — tagging every reported parameter with its provenance (measured / derived / estimated), its uncertainty, and the model version used. Table 2.3 is effectively the schema for that tagging.

Appendix A

Live Reference Library

This appendix collects every substantive source consulted for this review as a clickable link, organized by category. The four raw research dossiers in `tech-review/research/` pair each of these with the specific claims they support.

A.1 Patents (Google Patents / USPTO)

TrackMan A/S (Fredrik Tuxen)

- Full portfolio index: <https://patents.justia.com/inventor/fredrik-tuxen> and <https://www.trackman.com/legal/patents>
- US8845442 — Determination of spin parameters of a sports ball (harmonic-sideband spin; Magnus spin-axis inversion): <https://patents.google.com/patent/US8845442B2/en>
- US10393870 — continuation of the above: <https://patents.google.com/patent/US10393870B2/en>
- US10850179 — spin axis via receiver-pair interferometry: <https://patents.google.com/patent/US10850179B2/en>
- US8085188 / US9857459 / US10473778 — target-deviation (tap-a-target) family: <https://patents.google.com/patent/US8085188B2/en>, <https://patents.google.com/patent/US9857459B2/en>, <https://patents.google.com/patent/US10473778B2/en>
- US10315093 — trajectory tracer overlay: <https://patents.google.com/patent/US10315093B2/en>
- US11086005 — multi-bay range tracking: <https://patents.google.com/patent/US11086005B2/en>
- TrackMan’s own spin-patent explainer: <https://blog.trackmangolf.com/patent-measuring-the-spin-of-a-sport>

FlightScope / EDH (Henri Johnson)

- US9868044 — dielectric-lens spin measurement: <https://patents.google.com/patent/US9868044B2/en>
- US10775492 — direct spin-axis via perpendicular receiver pairs: <https://patents.google.com/patent/US10775492B2/en>
- US10338209 — Fusion Tracking (radar+camera): <https://patents.google.com/patent/US10338209B2/en>
- Company history: <https://athlonsports.com/golf/the-flightscope-story-tracking-the-history-of-flight>

Foresight Sports / Wintriss Engineering (Kiraly)

- US7292711 — flight parameter measurement system (**expired 2025**; markerless dimple-spin blueprint): <https://patents.google.com/patent/US7292711B2/en>
- US7324663 — smart-camera sibling (**expired 2025**): <https://patents.google.com/patent/US7324663B2/en>
- US7497780 — integrated launch monitor UX: <https://patents.google.com/patent/US7497780B2/en>
- US7641565 — ball-placement detection: <https://patents.google.com/patent/US7641565B2/en>
- WAWGD (Foresight) filings index: <https://patents.justia.com/inventor/wawgd-inc-dba-foresight-sports>
- Foresight–Uneekor 2024 license (Business Wire): <https://www.businesswire.com/news/home/20240930838753/en/>

Acushnet / Titleist (Gobush et al.)

- US5501463 — stereo + retroreflective dots, club and face (**expired**): <https://patents.google.com/patent/US5501463A/en>
- US6500073 — stereo ball trajectory + flight integration (**expired**): <https://patents.google.com/patent/US6500073B1/en>
- US6758759 — dual stereo monitors, measured face angle (**expired**): <https://patents.google.com/patent/US6758759B2/en>
- US7143639 — portable four-camera monitor (**expired**): <https://patents.google.com/patent/US7143639B2/en>
- US8500568 / US8556267 — portable continuations: <https://patents.google.com/patent/US8500568B2/en>, <https://patents.google.com/patent/US8556267B2/en>
- US10668350 — 3D/light-field launch monitor: <https://patents.google.com/patent/US10668350B2/en>
- US6186002 — C_D/C_L from measured trajectories (model calibration template): <https://patents.google.com/patent/US6186002B1/en>

Others

- Full Swing US2020/0147470 — CW+FMCW dual-mode launch monitor: <https://www.freepatentsonline.com/y2020/0147470.html>
- Creatz US10247553 — start-position sensor: <https://patents.google.com/patent/US10247553B2/en>
- Golfzon US9242158 — two-stage latency-hiding simulation: <https://patents.google.com/patent/US9242158B2/en>
- Uneekor patent/license list: <https://uneekor.com/legal/patents>
- Camera-timing/Gabor dimple tracking US12401909: <https://patents.google.com/patent/US12401909B2/en>
- TrackMan EP1698380 litigation coverage: <https://thegolfwire.com/322771-2/>
- FlightScope 2022 BGH win vs TrackMan: <https://golfbusinessnews.com/news/innovation-centre/flightscope-wins-patent-infringement-case-against-trackman-in-germany/>

A.2 Regulatory / teardown (RF ground truth)

- TrackMan 4 FCC ID SFX-TMAN4 (X-band + 24 GHz): <https://fccid.io/SFX-TMAN4>
- TrackMan iO FCC ID SFX-TMB0201 (24 GHz): <https://fccid.io/SFX-TMB0201>
- FlightScope Mevo FCC ID QXP-A7310 (24.125 GHz): <https://fccid.io/QXP-A7310>
- Mevo+ teardown / QXP-LJ361 (10.5–10.55 GHz) discussion: <https://golfsimulatorforum.com/forum/flightscope/277362-mevo-teardown>

A.3 Manufacturer technical documentation

TrackMan

- 40+ parameters explained: <https://www.trackman.com/blog/golf/40-trackman-parameters>
- Club data definitions: <https://www.trackman.com/blog/golf/club-data-definitions>
- Club speed (geometric-center definition): <https://www.trackman.com/blog/golf/what-is-club-speed>
- Spin loft / spin rate: <https://www.trackman.com/blog/golf/spin-loft>, <https://www.trackman.com/blog/golf/spin-rate>
- OERT (“Two radars, one camera, zero doubt”): <https://www.trackman.com/blog/golf/two-radars-one-camera-zero-doubt>
- Tech specs (TM4, iO): <https://www.trackman.com/golf/launch-monitors/tech-specs>
- Ball Flight Laws newsletter (Jan. 2009; D-plane, 85/15): <https://www.yumpu.com/en/document/view/11652756/trackman-ball-flight-laws>

Foresight Sports

- GCQuad product/technology: <https://www.foresightsports.com/products/gcquad-launch-monitor>
- Ball and club data (spherical correlation, fiducials): <https://foresightsports.eu/ball-club-data/>
- HMT head measurement (dot-count data tiers): <https://www.foresightsports.com/hmt-head-measurement>
- Club marker application guide: <https://help.foresightsports.com/hc/en-us/articles/4408197030035>
- Marker troubleshooting: <https://help.foresightsports.com/hc/en-us/articles/4405916455443>
- Design history (Popular Science): <https://www.popsci.com/gear/foresight-sports-quadmax-photometric-golf-club/>

Other vendors

- FlightScope X3 (multi-frequency, Fusion Tracking): <https://thegolfwire.com/326713-2/>
- Garmin R10 architecture and accuracy: <https://mygolfsimulator.com/garmin-r10-data/>, <https://mygolfsimulator.com/garmin-r10-accuracy/>
- Garmin R50 (three-camera): <https://www.garmin.com/en-US/p/736810/>
- Full Swing KIT technology and specs: <https://www.fullswinggolf.com/kit-launch-monitor-technology/>, <https://www.fullswinggolf.com/kit-specs/>
- Rapsodo MLM2PRO spin (MyGolfSpy): <https://mygolfspy.com/news-opinion/rapsodo-mlm2pro-takes-spin-measurements/>

- SkyTrak+ measurement: <https://www.skytrakgolf.com/pages/what-does-the-skytrak-st-launch-monitor-measure>
- Uneekor EYE XO / Dimple Optix: <https://uneekor.com/blogs/blog/photometric-vs.-doppler:-which-launch-monitor-technology-delivers-the-most-accurate-golf-data>
- Uneekor calibration guide (Carl's Place): <https://www.carlofet.com/blog/calibrating-uneekor-launch-monitors>
official PDF: https://download.uneekor.com/docs/EYEX02_Calibration_Guide.pdf
- ProTee VX reviews: <https://www.playbetter.com/blogs/golf-simulator-reviews/protee-vx-review>

A.4 Peer-reviewed and academic

- Penner, "The physics of golf," Rep. Prog. Phys. 66 (2003): <https://iopscience.iop.org/article/10.1088/0034-4885/66/2/202>
- Leach et al. 2017 launch-monitor validation (*Measurement*): <https://www.sciencedirect.com/science/article/abs/pii/S0263224117305079>; open-access: https://repository.lboro.ac.uk/articles/journal_contribution/9562799
- TrackMan 4 indoor reliability (J. Sports Sci. 2024): <https://www.tandfonline.com/doi/full/10.1080/02640414.2024.2314864>
- Mevo+ vs TrackMan 4 agreement (2025): <https://www.sciencedirect.com/science/article/pii/S2772696725000420>
- Bearman & Harvey 1976 golf ball aerodynamics: <https://www.cambridge.org/core/journals/aeronautical-quarterly/article/abs/golf-ball-aerodynamics/67FE0903DB1CC12001F1ED1B1261C4B9>
- Smits & Smith aerodynamic model: <https://www.researchgate.net/publication/284037213>
- USGA Indoor Test Range conditions: <https://www.usga.org/content/dam/usga/images/equipment-standards/ITR-test-conditions-2028-ODS.pdf>
- Friction/tangential compliance and launch angle (MDPI 2020): <https://doi.org/10.3390/proceedings2020049027>
- Face angle / club path influence study: <https://www.researchgate.net/publication/323373897>
- Cross oblique-impact golf experiments: <https://www.physics.usyd.edu.au/~cross/GOLF/GOLF.htm>
- Cross & Nathan oblique collision formalism: <https://arxiv.org/pdf/1610.03464>
- Dynamic models review, Sports Engineering 25 (2022): <https://link.springer.com/article/10.1007/s12283-022-00387-0>
- Stanford LES of golf ball flow (Sports Eng. 2019): http://aero-comlab.stanford.edu/Papers/golf_ball_sports_engineering_2019.pdf
- Golf ball aerodynamics in still air (MDPI): <https://www.mdpi.com/2504-3900/2/6/238>
- Micro-Doppler of spinning projectiles on CW radar: <https://www.researchgate.net/publication/317119893>
- SpinDOE camera spin estimation (transferable method): <https://arxiv.org/pdf/2303.03879>
- "Measuring Ball Spin by Image Registration": <https://www.researchgate.net/publication/2881136>
- Doppler ball-tracking thesis (WSU, TrackMan platform): <https://baseball.physics.illinois.edu/trackman/JasonMartinThesisWSU.pdf>

- Alan Nathan’s TrackMan (baseball) technical notes: <https://baseball.physics.illinois.edu/trackman.html>
- Cochran & Stobbs, *Search for the Perfect Swing* (1968): <https://archive.org/details/searchforperfect0000coch>

A.5 Engineering references (Tutelman et al.)

- 3D launch conditions from impact conditions: <https://www.tutelman.com/golf/ballflight/3dlaunch.php>
- Smash factor: <https://www.tutelman.com/golf/ballflight/smashfactor.php>
- Gear effect: <https://www.tutelman.com/golf/ballflight/gearEffect1.php>
- Spin decay: <https://www.tutelman.com/golf/ballflight/spinDecay.php>
- Ball flight laws summary: <https://www.perfectgolfswingreview.net/ballflight.htm>
- TrackMan face-angle derivation discussion (Manzella forum): <https://forum.brianmanzellagolf.com/threads/how-does-trackman-flightscope-measure-the-club-face-angle.16591/>

A.6 DIY / open-source

- PiTrac GitHub (GPL-2.0): <https://github.com/PiTracLM/PiTrac>
- PiTrac documentation: <https://docs.pitrac.org/>
- PiTrac Hackaday.io project (design logs): <https://hackaday.io/project/195042-pitrac-the-diy-golf-launch-mo>
- OpenFlight upstream: <https://github.com/jewbetcha/openflight>
- OpenFlight Grafana write-up: <https://medium.com/grafana-labs/openflight-building-an-open-source-golf-l>
- GC2-imitating OpenCV experiments: <https://github.com/ronheywood/opencv>
- GSA Golf simulator theory (marked-ball DIY methods): <https://www.golf-simulators.com/GolfSimulatorTheory.html>

A.7 Comparative testing and market analysis

- MyGolfSpy indoor accuracy test: <https://mygolfspy.com/news-opinion/three-of-the-most-accurate-indoor-l>
- GCQuad vs TrackMan robot spin data: <https://golfsimulatorzone.com/gcquad-vs-trackman-accuracy-in-2026/>
- Foresight GCQuad/GC3 comparison: <https://www.foresightsports.com/blogs/golf-tips/comparing-golf-laun>
- Bushnell Launch Pro vs GC3 vs GCQuad: <https://www.playbetter.com/blogs/golf/bushnell-launch-pro-vs-gc3>
- Photometric vs radar explainers: <https://www.playbetter.com/blogs/golf-simulator-comparisons/photometric-vs-radar-golf-launch-monitor>, <https://golfbays.de/en/blogs/news/radar-vs-photometric-l>
- Indoor flight compensation: <https://www.cerogolf.com/post/how-launch-monitors-compensate-for-indoor-bal>
- TrackMan history (Golf Monthly, Tuxen profile): <https://www.golfmonthly.com/features/from-tracking-missile>

Appendix B

Detailed Implementation Guidance for OpenFlight

This appendix turns the review into engineering guidance at the level of signal-processing parameters, algorithms, and procedures, citing the governing sources throughout. Coordinate conventions follow Chapter 2.

B.1 Radar signal chain (OPS243-A)

B.1.1 Doppler scaling and resolution budget

At $f_c = 24.125$ GHz, Equation (4.1) gives 71.7 Hz per mph. With OpenFlight’s 30 kHz I/Q sample rate, the unambiguous span is ± 15 kHz $\approx \pm 209$ mph — adequate for ball speeds to ~ 200 mph. Velocity resolution is set by observation time: a 128-sample window (4.27 ms) gives $\Delta f = 234$ Hz ≈ 3.3 mph per raw bin; zero-padding to 4096 (the current pipeline) interpolates the peak but does not add information. For *spin* work the window, not the padding, must grow: resolving sidebands at $f_{\text{spin}} = S/60$ (e.g. 50 Hz at 3,000 rpm) requires windows of $\gtrsim 40$ ms, during which a 150 mph ball decelerates and the central line *chirps* — so the practical estimator de-chirps first (track the central line per US8845442’s spectral-trace step [13], resample the phase to remove it), then measures the residual modulation.

B.1.2 Spin-rate comb estimation

The patent-documented chain (Section 4.4.2) maps to this concrete pipeline [13, 36]:

1. STFT with 50–75% overlap over the post-impact 50–250 ms; track the ball ridge (max-SNR bin per frame with continuity constraint).
2. De-chirp: mix each frame down by the tracked ridge frequency so the ball line sits at DC.
3. Estimate the sideband comb spacing on the de-chirped spectrum via cepstrum or harmonic product spectrum — both are robust to missing harmonics; the patent’s *qualification* step (verify equal spacing across ≥ 3 consecutive frames, and consistency of the harmonic-number assignment) is what separates real spin from clutter [13].
4. Report spin only when the qualified comb persists; otherwise fall back to a flagged estimate from club priors (Section B.3) — the Garmin R10’s documented behavior [43].

Expected detection physics: sideband amplitude scales with ball-surface asymmetry [13]; range balls and clean urethane balls read weakly (the reason for Titleist RCT metal-tagged balls [46] and FlightScope’s metallic-dot stickers [40]). Logging which ball type produced each detection will quantify this for OpenFlight’s own statistics.

Patent posture: this method is claimed by US8845442 to ~2029 (US10393870 to Dec. 2026) [13]; the FlightScope phase-demodulation alternative (US9868044, dielectric lens [16]) is claimed to ~2034. See Section 7.6.

B.1.3 Club-speed extraction

The pre-impact spectrum contains a velocity *smear* from heel (slow) to toe (fast), plus shaft returns below. TrackMan resolves this by reconstructing the head silhouette and reporting the geometric center [33]; a single-radar approximation that tracks a stable reference is: gate the last 30–50 ms before the trigger timestamp, form the velocity histogram of CFAR-passing bins, discard the bottom decile (shaft/hosel) and top decile (toe glint), and report a fixed percentile (median of the remainder). Validate against the loft-dependent smash ceiling of Section 3.2 ([59]); reject or flag shots exceeding it, which independent testing shows is the signature of toe-lock errors [7].

B.2 Angle measurement (K-LD7) and the EKF

B.2.1 Interferometric angles

The K-LD7’s two receive patches implement Equation (4.2) with a $\lambda/2$ -class baseline; per-bin phase comparison of the two ADC channels after the range-Doppler FFT yields one angle per detection — the same phase-monopulse principle as US10850179 [14] and the R10’s three-receiver array [42]. Two practical constraints from the commercial art: (i) angle error grows as SNR falls, so weight each detection by measured SNR; (ii) mechanical alignment dominates the error budget — a 1° mount error is a 1° bias on every launch direction, amplified $\sim 1.15\times$ into face angle (Equation (4.6)).

B.2.2 Recommended filter

Implement the Section 8.3 smoother concretely as: state $\mathbf{x} = (\mathbf{p}, \mathbf{v})$ (add spin states later); process model Equation (8.1) with Smits–Smith coefficients [5]; measurements: OPS243-A radial speed ($\dot{r}$, high rate, low noise), K-LD7 angle+range detections (lower rate, SNR-weighted covariance). Run a forward EKF then an RTS (Rauch–Tung–Striebel) backward smoother over the 50–150 ms burst, and evaluate the smoothed state at the sound-trigger timestamp minus the acoustic delay — this back-extrapolation is how commercial radars report launch conditions robustly despite early-flight clutter [13, 18]. The innovation-gating step of the EKF doubles as the outlier rejector for multipath and club returns.

B.2.3 Alignment calibration procedure

Adopt the commercial patterns: (i) Unekor-style floor chart — a printed target-line chart at known positions establishes the target-line azimuth for both K-LD7s [55]; (ii) FlightScope-style known-trajectory check — roll or swing balls along a surveyed line and verify reported directions (US10338209 uses a Doppler simulator for the same purpose [18]); (iii) record the mount pose in the session log so data from different setups is never silently mixed.

B.3 Derived club data, with priors

Implement the D-plane module once, use it both ways (Sections 3.1 and 4.5.2):

- **Forward** (simulation/validation): Tutelman’s calibrated closed form, Equations (3.3) and (3.4), plus spin from Equation (3.7) [58].
- **Inverse** (reporting): face angle from Equation (4.6) with w_f interpolated in dynamic loft between 0.87 (driver) and 0.75 (wedge) [35, 58]; dynamic loft analogously from launch angle and attack angle. Tag both as *derived* per Table 2.3.
- **Priors table** per club type (driver...wedge): static loft, typical dynamic-loft delta, spin-loft range, smash ceiling (Section 3.2), PGA/LPGA reference deliveries [31]. Used for: spin fallback estimates, outlier gating, and the smash sanity check.
- **Gear-effect bound**: when impact location is unknown, attach an uncertainty of up to $\pm 6^\circ$ of spin-axis tilt per 0.14in of possible driver miss [61, 60] to any derived face-to-path interpretation — and surface it in the UI rather than hiding it.

B.4 Optical module (Phase 2) design parameters

The expired Wintriss patents [19] plus PiTrac’s published design [66] fix the working parameters:

- **Sensor**: Raspberry Pi Global Shutter camera (IMX296, 1456×1088); global shutter is non-negotiable (rolling shutter skews a 150mph ball by several pixels per row-time).
- **Strobed multi-exposure**: $N = 3\text{--}5$ IR pulses (850nm, tens of μs each) inside one long exposure freeze N ball images per frame; at 150mph the ball moves 6.7cm/ms, so pulse spacing of $\sim 300\text{--}500\mu\text{s}$ spaces images 2–3cm apart in a 30cm capture volume — matching the commercial capture geometry [47, 66].
- **Trigger**: reuse the SEN-14262 sound trigger; its $\sim 10\mu\text{s}$ latency is far inside the strobe-timing budget, solving the triggering problem the Wintriss patent addressed with a microphone+radar pair [19].
- **Ball detection**: Hough circles on the strobed frame (PiTrac-proven [66]); depth from the known 42.67mm diameter (the Wintriss mono-camera range cue [19]) or from a second camera via Equation (5.1).
- **Spin**: register the dimple texture between successive ball images over $SO(3)$ (Equation (5.2)); Gabor-filter pre-enhancement of dimples is documented in US12401909 [26]; with $\Delta t \approx 400\mu\text{s}$, 3,000rpm is only 7.2° of rotation — comfortably below the aliasing limit, and small enough that a local search around the D-plane-predicted axis converges quickly (Section 5.2.2). This is the public-domain (post-2025) Wintriss/Foresight method [19, 50].
- **Impact location / measured face angle** (later): either face fiducials per the expired Acushnet recipes (US5501463/US6758759: dots + stereo pose) [20, 22], or an overhead second camera per the Uneekor geometry [52].

B.5 Flight model and validation protocol

Ship Smits–Smith (Equation (8.2)) with spin decay [5], version the coefficients, and calibrate against measured trajectories using the US6186002 fitting approach [23] — outdoor sessions with the MLM2PRO (or simple carry ground-truth) provide the data. Validation against the MLM2PRO should follow the Leach protocol structure [7]: per-club shot blocks, Bland–Altman limits of agreement per parameter, ball data compared directly, spin compared only on RPT/RCT balls

(where the MLM2PRO’s spin is camera-measured [46]), and club speed compared with an explicit reference-point caveat (Section 4.5). Log raw I/Q and (later) raw frames for every shot so algorithm changes can be replayed against history — the practice that made this review’s accuracy analysis possible for the commercial units is exactly what an open project can do better.

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